KiranAcademy
IT Training & Placement

Python Dictionary Practice Tasks

Basic, Logical, Nested Dictionary and Challenging Tasks

Basic Dictionary Tasks

Use this dictionary for Tasks 1-15:

```
student = {
    "name": "Rahul",
    "age": 22,
    "course": "Python",
    "city": "Pune",
    "marks": 78
}
```

1. Print the complete dictionary.
2. Print only the student's name.
3. Print the value of the course key.
4. Display all dictionary keys.
5. Display all dictionary values.
6. Display all key-value pairs.
7. Find the total number of key-value pairs.
8. Check whether the key "city" exists.
9. Add a new key "email" with any email address.
10. Update the marks from 78 to 85.
11. Change the city from Pune to Mumbai.
12. Remove the age key from the dictionary.
13. Use get() to display the value of the phone key without causing an error.
14. Create a copy of the dictionary.
15. Clear all elements from the copied dictionary.

Logical and Nested Dictionary Tasks

Use this dictionary for Tasks 16-30:

```
employees = {
    101: {"name": "Amit", "department": "IT", "salary": 45000},
    102: {"name": "Priya", "department": "HR", "salary": 38000},
    103: {"name": "Sneha", "department": "IT", "salary": 52000},
    104: {"name": "Raj", "department": "Sales", "salary": 41000},
    105: {"name": "Anjali", "department": "HR", "salary": 47000}
}
```

16. Print the name of every employee.
17. Print employee ID with employee name.
18. Print employees who belong to the IT department.
19. Print employees whose salary is greater than 40,000.
20. Find the employee with the highest salary.
21. Find the employee with the lowest salary.

22. Calculate the total salary of all employees.
23. Calculate the average salary of all employees.
24. Count how many employees are present in each department.
25. Increase the salary of every employee by 10%.
26. Create a new dictionary containing employee names and salaries.
27. Create a new dictionary containing only HR employees.
28. Find whether employee ID 103 exists.
29. Remove employee ID 104 from the dictionary.
30. Add a new employee with ID 106.

Product Dictionary Tasks

Use this dictionary for Tasks 31-40:

```
products = {
    "P101": {"name": "Laptop", "price": 55000, "stock": 8},
    "P102": {"name": "Mouse", "price": 700, "stock": 25},
    "P103": {"name": "Keyboard", "price": 1200, "stock": 15},
    "P104": {"name": "Monitor", "price": 14000, "stock": 5},
    "P105": {"name": "Headphones", "price": 2500, "stock": 0}
}
```

31. Print the name and price of every product.
32. Print products having stock less than 10.
33. Print products that are out of stock.
34. Calculate the total value of each product using price x stock.
35. Find the most expensive product.
36. Find the cheapest product.
37. Create a list containing the names of all available products.
38. Create a dictionary containing product name as key and stock as value.
39. Update the stock of product P104 to 12.
40. Apply a 5% discount to all products costing more than 10,000.

Challenging Dictionary Tasks

41. Merge two dictionaries without using the | operator.
42. Find common keys between two dictionaries.
43. Find keys present in the first dictionary but not in the second.
44. Create a dictionary from two separate lists: one containing keys and another containing values.
45. Reverse a dictionary so that values become keys and keys become values.
46. Count the frequency of every character in a user-given string using a dictionary.
47. Count the frequency of every word in a sentence.
48. Group student names by their city using a dictionary.
49. Sort a dictionary by its values in ascending order.
50. Remove duplicate values from a dictionary while keeping the first matching key.

Total Tasks: 50	Topic: Python Dictionary	Answers: Not Included